

AR Episodic Memory Glasses

Wearable glasses that retrieve and replay relevant moments from a recent first-person video and audio history.

Need: People often forget details of recent conversations and activities. Existing recordings are difficult to search when the user remembers the meaning of an event but not when it occurred.

Initial users: Students and professionals who need to recover recent instructions, decisions and interactions. Support for people with memory impairments is a possible future application that would require appropriate clinical input.

Proposed solution: AR glasses capture first-person video and audio during enabled periods. A companion device segments and indexes the recordings. The user asks a natural-language question, receives a grounded answer, and can replay the supporting footage through the display.

Core Scope

- Capture synchronized first-person video and audio during user-enabled periods.
- Create timestamped transcripts and searchable descriptions of short video segments.
- Retrieve and rank relevant events from vague natural-language queries.
- Present a grounded answer and replay its original clip through AR.

Constraints and Risks

- At roughly 2 Mb/s, twelve hours of video creates about 10.8 GB per day and 76 GB per week.
- Continuous capture stresses battery life, thermal management, comfort and wireless transfer.
- Recording other people creates consent, privacy and social-acceptance concerns.
- Poor lighting, obstruction and motion blur can leave important moments uncaptured.

Preliminary Success Measures

Measure	Preliminary target
Event retrieval	Correct event appears in the top three for at least 80 percent of a defined test set
Response time	Relevant result returned within approximately 10 seconds for the test archive
Evidence	Every factual answer links to the original synchronized clip
Retention	Expired video, audio, transcript and index data are deleted after seven days

Mechatronics coverage Glasses packaging and thermal design; cameras, microphones, battery and display electronics; embedded capture and storage; multimodal ML retrieval; companion software and AR playback.

Faculty feedback requested Is this sufficiently distinct from Rekall and Recall? Is several-hour capture with a seven-day archive a defensible MVP? Would integrating an existing AR display provide enough hardware depth?

SmokeScout Drone

A compact indoor drone that maps a smoke-obscured test environment and reports thermal targets and major hotspots before responder entry.

Need: Firefighters entering smoke-obscured buildings may have limited information about occupants, obstacles and major heat sources. A small recon platform could provide an initial map before personnel enter.

Initial users: Firefighters and emergency-response teams performing pre-entry assessment of enclosed structures.

Proposed solution: Build a protected indoor drone that localizes without GPS, maps a controlled environment, detects heated human-sized targets and hotspots, and sends their locations to an operator station.

Core Scope

- Design and integrate a compact airframe, guards, sensor payload and power system.
- Perform indoor localization, mapping and obstacle avoidance without GPS.
- Detect and locate controlled thermal human targets and major hotspots.
- Replan when the current route becomes blocked.
- Transmit the map and detections and execute lost-link and low-battery failsafes.

Validation Boundary

Testing would use theatrical smoke, heated mannequins or controlled heat sources, and a configurable obstacle course. The prototype would not be tested in a burning building, suppress fire, or claim emergency-service certification.

Constraints and Risks

- Smoke, featureless surfaces and changing visibility may degrade localization sensors.
- Propeller wash can disturb smoke and lightweight debris, while walls can degrade communications.
- The payload reduces flight time and increases control complexity.
- Detecting and avoiding falling debris fast enough is likely beyond the MVP.
- Developing the airframe, flight controls, SLAM, planning and thermal perception together may exceed eight months.

Preliminary Success Measures

Measure	Preliminary target
Navigation	Complete the defined smoke course without collision in at least 8 of 10 trials
Target detection	Meet a defined detection-rate target for heated human-sized targets
Localization	Report target position within approximately 0.5 to 1 metre in the test course
Replanning	Recover when a known route becomes blocked
Failsafe	Land safely or return after lost communication or low battery

Mechatronics coverage Mechanical airframe and guards; power and sensing electronics; embedded flight control; SLAM and thermal perception; path planning; closed-loop motion control.

Faculty feedback requested Is a controlled smoke-filled changing-obstacle course acceptable validation? Should falling-debris avoidance be excluded? Is building the drone platform and autonomy stack too broad for this team?

DriveSight Portable Lane Aware HUD

A portable dashboard HUD that combines smartphone route guidance with camera-based ego-lane estimation.

Need: Drivers of older or lower-trim vehicles often look toward a phone or centre display for navigation. Low-cost aftermarket HUDs may show speed and turn instructions but generally do not understand the vehicle's actual lane.

Initial users: Drivers of vehicles without an integrated HUD or lane-aware navigation display.

Proposed solution: A phone supplies the upcoming maneuver and recommended lanes. A rigidly mounted forward camera estimates the current lane. The device combines these states and displays concise prompts through a reflected windshield HUD.

Core Scope

- Receive maneuver, distance, road, speed and recommended-lane data from a phone.
- Build a dedicated high-brightness reflected HUD with adaptive brightness.
- Estimate visible lane boundaries and the vehicle's current lane from a forward camera.
- Fuse route intent and ego-lane state into advisory visual and audio prompts.
- Suppress lane-specific guidance when perception confidence is low.
- Install without permanent vehicle modification or CAN-bus access.

Scope Boundaries

The core prototype will not control the vehicle, certify that a lane change is safe, or require CAN access. Full road-registered AR remains outside the baseline scope.

Stretch scope: If the LCD-and-combiner implementation does not provide sufficient technical depth, add limited registered AR that anchors lane guidance to detected road geometry. This extension would introduce camera-to-display calibration, perspective projection, latency compensation and registration-error testing, without expanding to a full production-grade AR windshield.

Constraints and Risks

- An affordable display and combiner may be difficult to read in daylight without glare.
- Lane perception must handle faded markings, curves, shadows, night conditions and occlusion.
- Camera calibration must tolerate different dashboard positions and vehicle geometries.
- Required navigation and lane data must be available in Canada through the selected SDK configuration.
- The mount and information layout must avoid obstructing or distracting the driver.

Preliminary Success Measures

Measure	Preliminary target
Navigation interface	More than 95 percent of logged updates parsed correctly on controlled routes
HUD latency	Perception-driven changes displayed within approximately 250 milliseconds
Guidance logic	More than 95 percent agreement with labelled route and lane states in replay tests
Lane perception	Accuracy reported by road and lighting condition on withheld routes
Readability	Core information passes a defined viewing-distance and illumination test
Reliability	Complete a representative trip or replay session without crash or disconnect

Mechatronics coverage HUD optics and packaging; display, camera and power electronics; embedded computation and communication; computer vision, fusion logic and driver-interface design.

Faculty feedback requested Is fusion of route-recommended lanes with camera-estimated ego lane a strong enough contribution? What level of road testing and human-factors evidence would be expected?